Agile-Git-Maven

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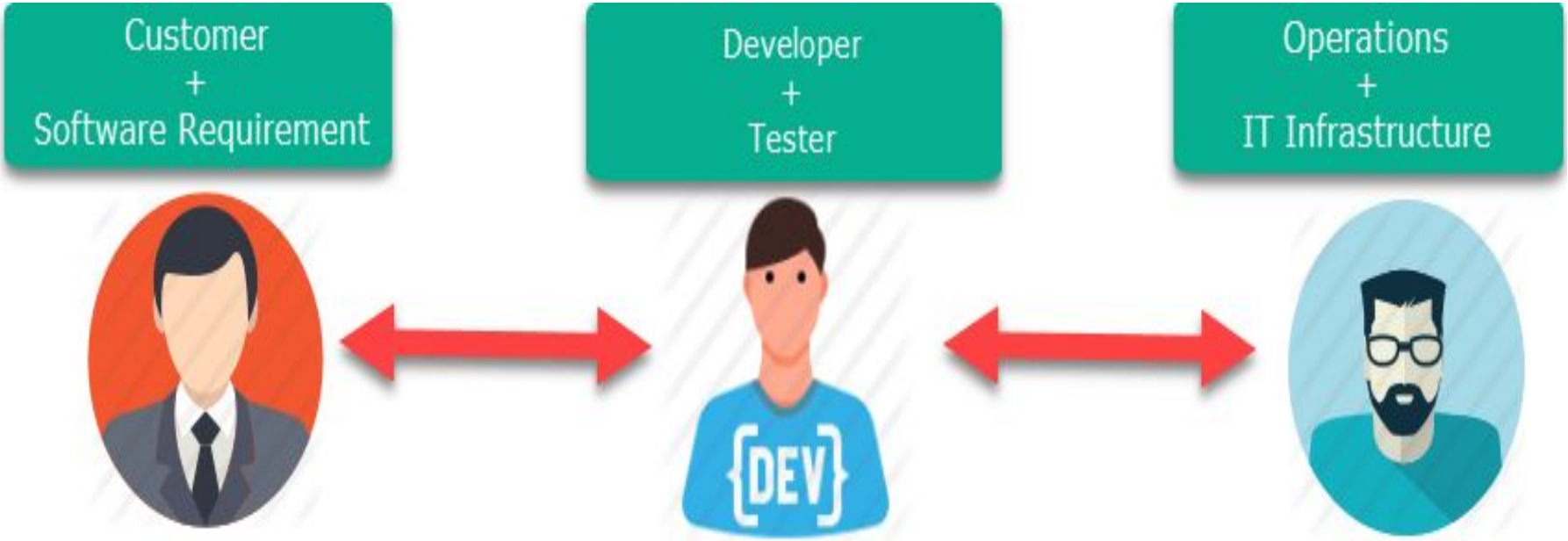
LEARNING OBJECTIVES

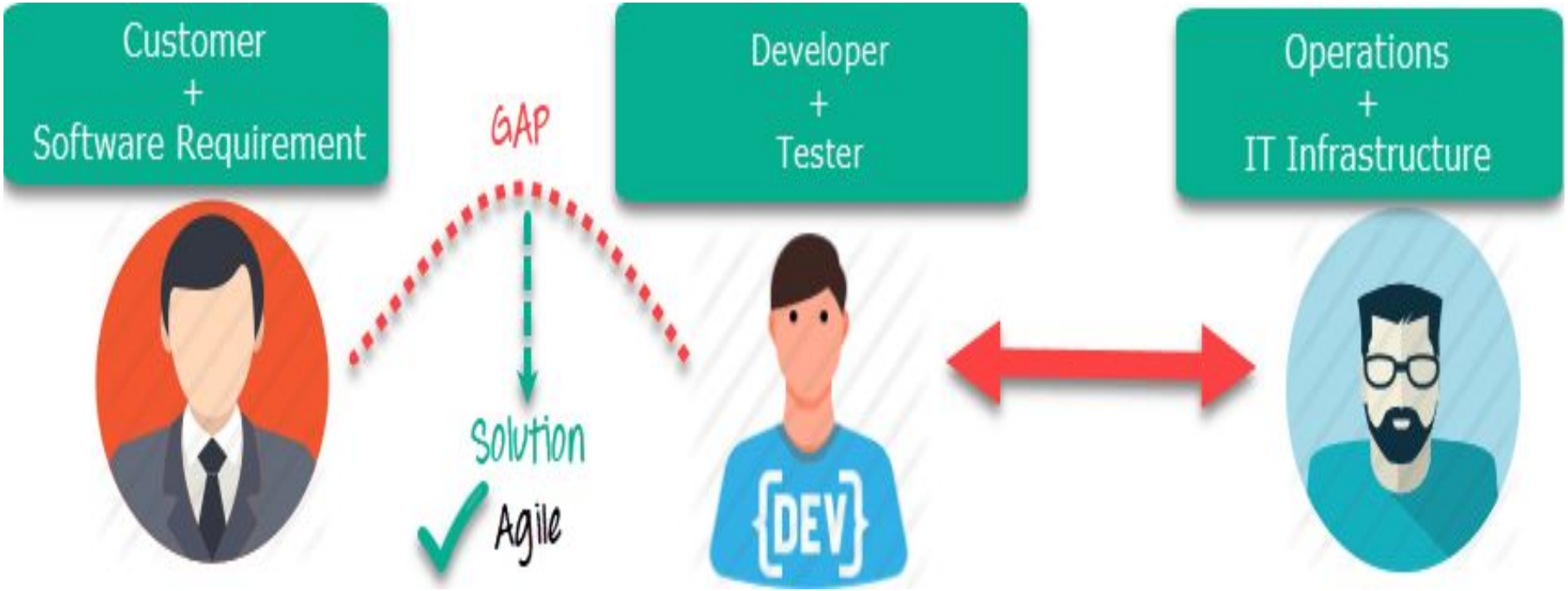
At the end of this lesson, you will be able to:

- Introduction to Agile
- Software Development

Software Development



Agile Advantage



CONCEPT

Scrum Basics

What is Scrum?

- Scrum refers to a holistic or “rugby” approach— where teams go the distance as a unit, passing the ball back and forth—as opposed to the traditional sequential or “relay race” approach for managing new product development.
- It is not individual effort but the best coordination of the team that will determine success.

User Stories: Components

- A User Story describes functionality that will be useful to a stakeholder of the system.

User Stories consists of three things

1. A brief description of the story used for
2. Conversations about the story
3. Tests that convey and document details
4. Assumptions
5. Estimating elements

IBM Customized Additional Elements

An example of user stories

User Story

Success would be …

- Acceptance Test #1
- Acceptance Test #2

As a system admin at Wal*Mart, I want to configure WAS so I can manage my applications

Success would be …

- WAS is configured
- No critical errors occurred in the process
- No data was lost

What is an Epic?

- As a premium site member, I want to cancel a reservation on the last minute with no cancellation fee so that I am not charged for a travel arrangement I no longer need.

EPIC

As an user, I want to cancel a reservation so that I am not charged for a travel arrangement I no longer need.

User Story

As a premium site member, I want to cancel a reservation on the last minute with no cancellation fee so that I am not charged for a travel arrangement I no longer need.

As a non-premium member, I want to cancel 24 hours in advance so that I am not charged for a travel arrangement I no longer need.

As a site visitor,

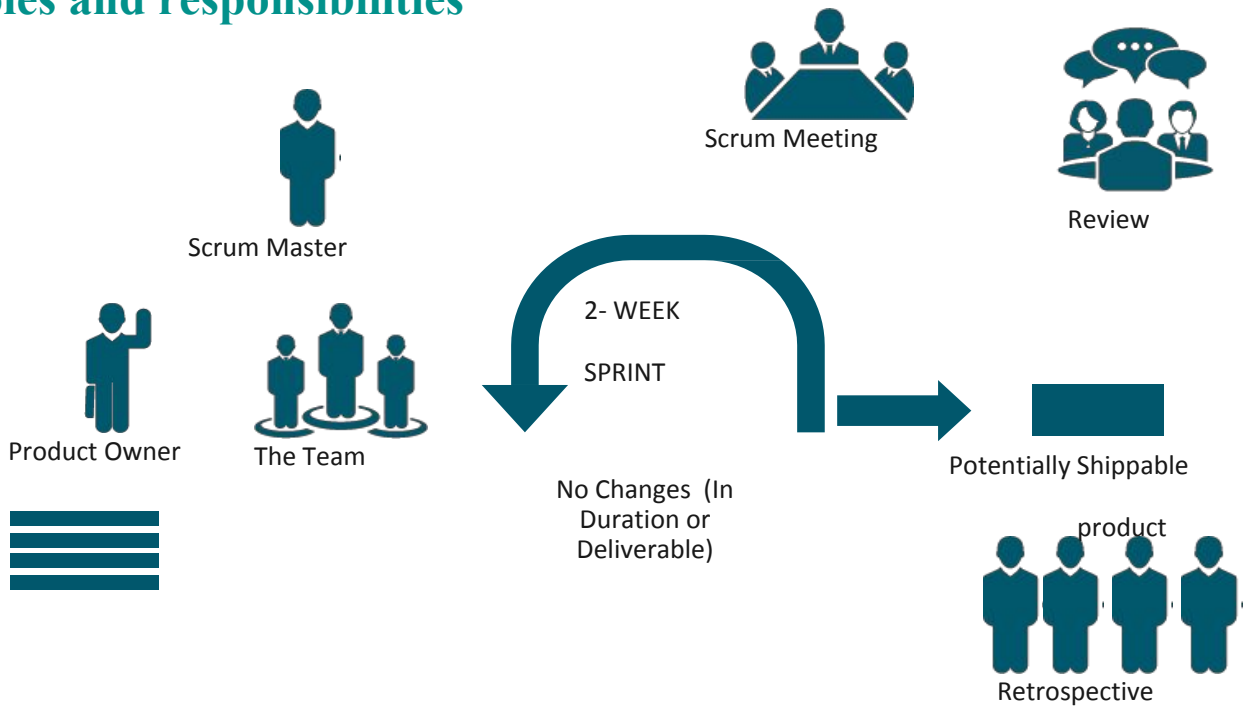
I want to receive a confirmation of any cancelled reservation so that I can have proof of cancellations.

Product Backlog

To-Do List

ID	Story	Estimation	Priority
7	As an unauthorized user, I want to create a New account	3	1
1	As an unauthorized user, I want to login	1	2
10	As an authorized user, I want to logout	1	3
9	Create script to purge database	1	4
2	As an authorized user, I want to see the list of items so that I can select one	2	5
4	As an authorized user, I want to add a new item so that it appears in the list	5	6
3	As an authorized user, I want to delete the selected item	2	7
5	As an authorized user, I want to edit the selected item	5	8
6	As an authorized user, I want to set a reminder for a selected item so that I am reminded when item is due	8	9
8	As an administrator, I want to see the list of accounts on login	2	10
Total		30	

Scrum roles and responsibilities



Scrum roles and responsibilities—product owner

Scrum roles and responsibilities—the team

7 people + or – 2

- Can be shared with other teams (but it is better when not)
- Can change between Sprints (but better when they don't)
- Can be distributed (but better when collocated)

Cross-functional

- Possesses all the skills necessary to produce an increment of potentially shippable product
- Team takes on tasks based on skills, not just the official "role".

Self-managing

- Team manages itself to achieve the Sprint commitment.

Scrum Master Roles

Scrum Master

Serves the team

The Scrum Master takes action to help remove impediments to the team's effectiveness.

The Scrum Master facilitates the team's group interactions to help the team achieve its full potential.

Protects the Team

The Scrum Master protects the team from anything that threatens its effectiveness, such as outside interference or disruption.

The Scrum Master will need to confront uncomfortable issues, both inside and outside the team.

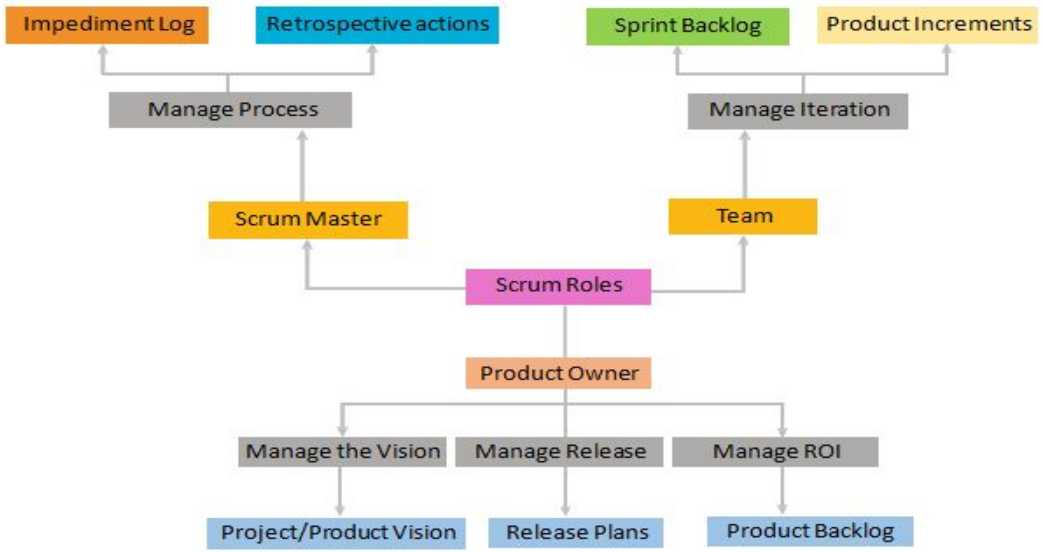
Guides the team's use of Scrum

Teaches Scrum to the team and

The Scrum Master ensures that all standard Scrum rules and practices are followed.

The Scrum Master organizes all Scrum- related practices.

Scrum project roles in a nutshell



Sprint planning

What is on the Product Owner's "shopping list"?

Team understands details of what Product Owner has prioritized on Product Backlog.

How much "money" do we have in our bank account?

Team decides how much productive time is available during the Sprint.

How many items on the shopping list can we afford to "buy" with that "monev"?

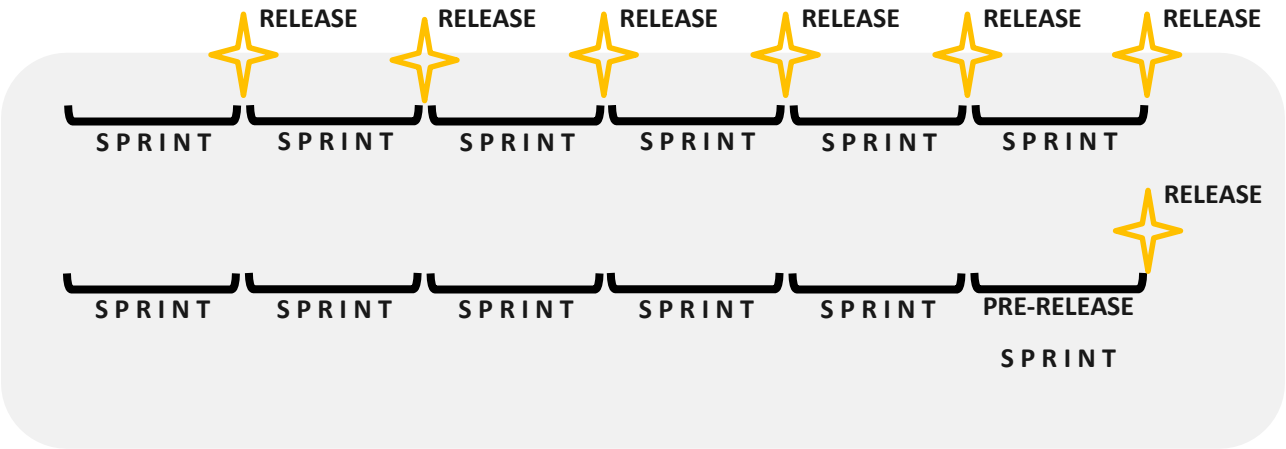
Team decides how many Product Backlog items can be committed for completion during the Sprint.

Sprint Backlog

			Day of Sprint						
Backlog Item	Task	Owner	Initial Est.	1	2	3	4	5	6
Enable all users to place book in shopping cart	Design business logic	Sanjay	4						
	Design user interface	Jing	2						
	Implement back-end code	Tracy	2						
	Implement front-end code	Tracy	6						
	Complete documentation	Joe	8						
	Unit testing	Philip	4						
	Regression testing	Philip	2						
Upgrade transaction processing module	Implement back-end code	Tracy	5						
	Complete documentation	Joe	6						
	Unit testing	Philip	3						
	Regression testing	Philip	3						
Total			214						

Definition of Done

In Scrum, “Done” is defined as “**Potentially Shippable**”.



CONCEPT

Scrum Execution

Daily Scrum Meeting

- Every weekday
- Whole team attends
- Everyone stands
- Lasts 15 minutes or less
- Product Owner can attend and report
- Update of artifacts after standup-Sprint backlog, Task board, and so on

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Daily Scrum Meeting (continued)

Every Developer / Tester reports the following three things:

What task was I able to accomplish since yesterday?

What task will I try to accomplish for the day?

Any issues blocking my work?

No discussions or conversations until meeting ends

Conduct Daily Scrum Meeting in the following scenario:

- There are three developers and two testers working on building a product that will automate the Bank Transactions.
  - One person is working on Creating new accounts.
  - Sub-tasks:
  - Create a Registration/login page.
  - Add options for User to select Creating new account.
  - Add all the mandatory / optional fields as text box or combo boxes.
  - Validate the data entered and the formats and save the data in the database.

Conduct Daily Scrum Meeting in the following scenario:

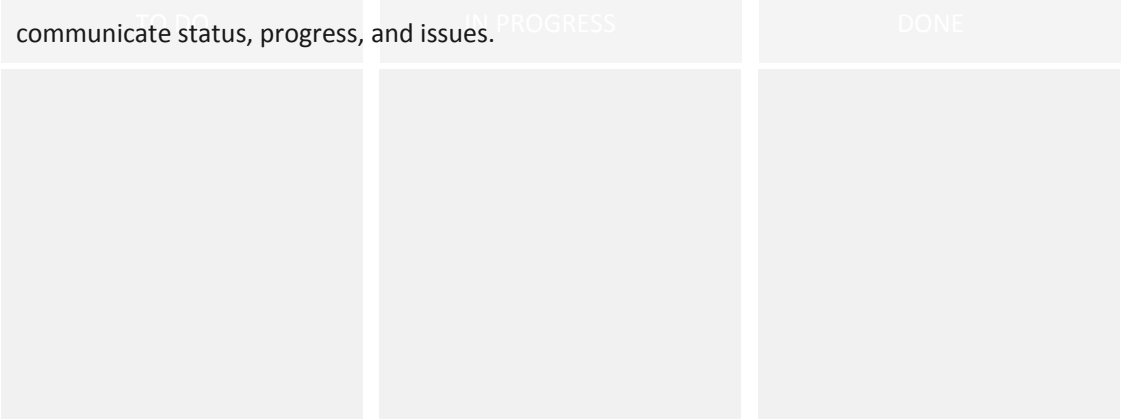
- Design a screen/page to provide the Report generating options.
  - Create the tables and views.
  - Design the report layout for different queries (ex: fetch all the details of Customer XYZ like account summary, FD details, Home loan details and so on.
- 2 Testers testing first 2 scenarios.

Sprint Backlog Updation

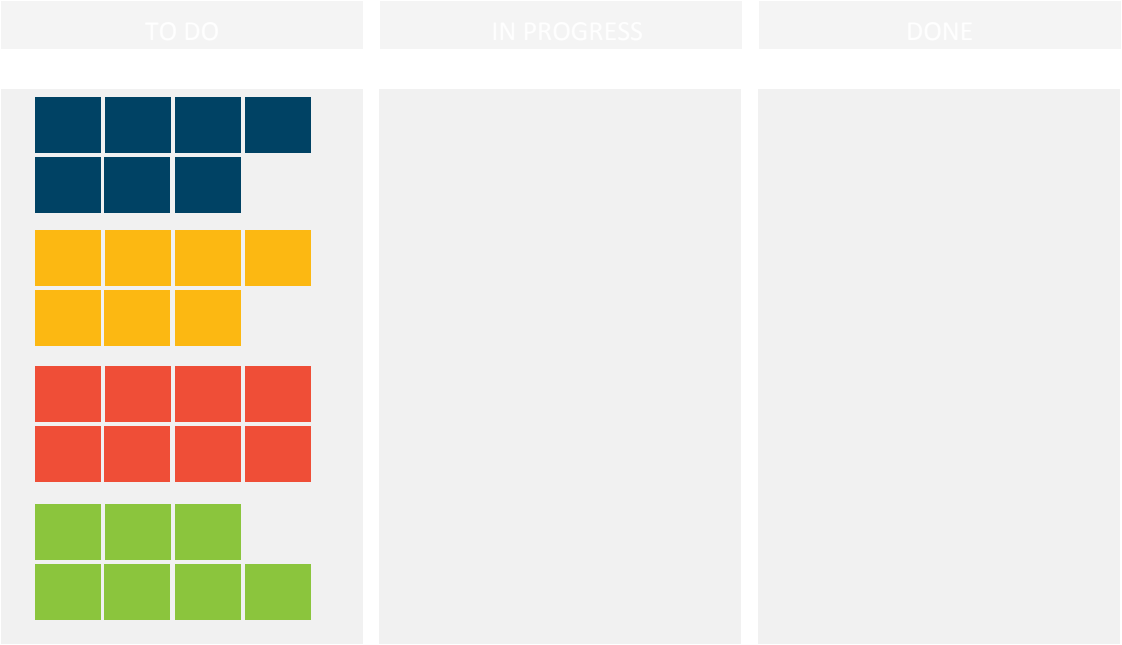
			Day of Sprint						
Backlog Item	Task	Owner	Initial Est.	1	2	3	4	5	6
Enable all users to place book in shopping cart	Design business logic	Sanjay	4	2	0				
	Design user interface	Jing	2	2	2				
	Implement back-end code	Tracy	2	4	2				
	Implement front-end code	Tracy	6	6	6				
	Complete documentation	Joe	8	6	6				
	Unit testing	Philip	4	3	3				
	Regression testing	Philip	2	2	2				
Upgrade transaction processing module	Implement back-end code	Tracy	5	10	8				
	Complete documentation	Joe	6	6	6				
	Unit testing	Philip	3	3	3				
	Regression testing	Philip	3	2	2				
Total			214	220	205				

Kanban Task Board (1 of 4)

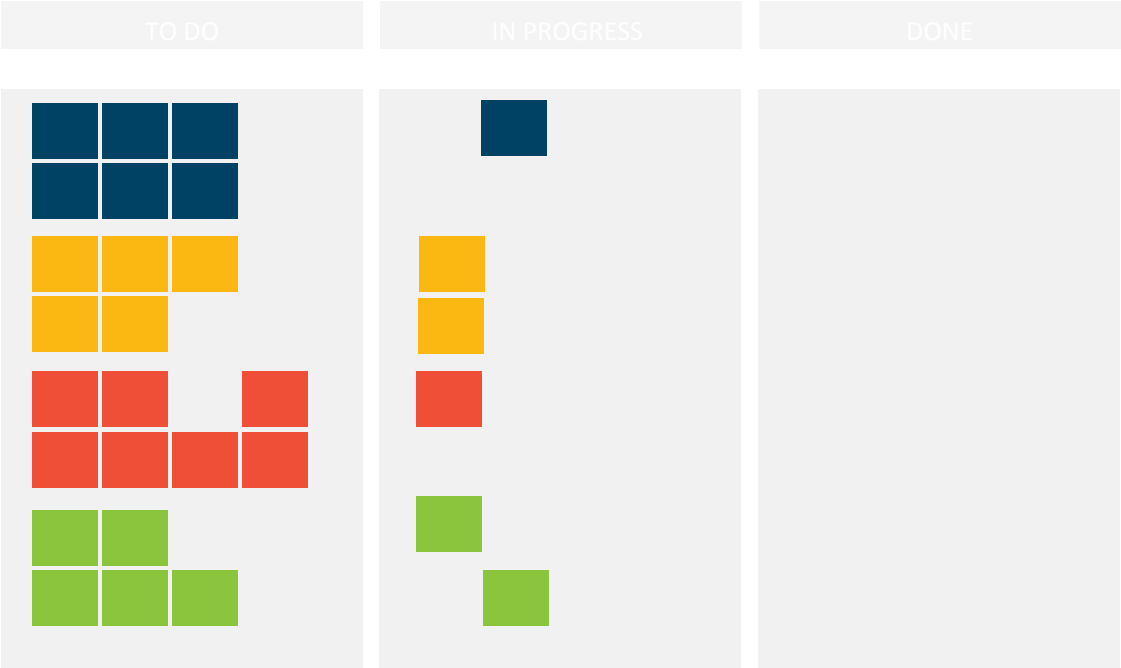
A Kanban board is a work and workflow visualization tool that enables you to optimize the flow of your work. Physical Kanban boards, like the one pictured below, typically use sticky notes on a whiteboard to communicate status, progress, and issues.



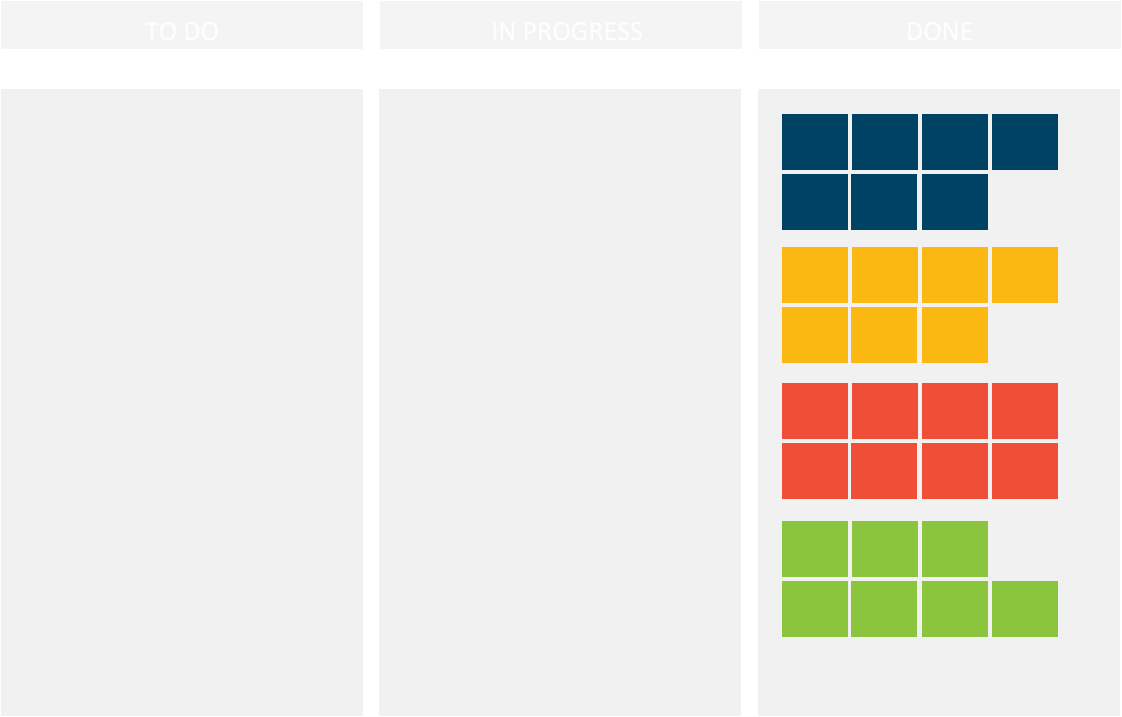
Kanban Task Board (2 of 4)



Kanban Task Board (3 of 4)



Kanban Task Board (4 of 4)



CONCEPT

Sprint Review and Retrospective

Sprint Review

- The purpose of the Sprint Review is:
  - Demo what the team has built
  - Generate feedback which the Product Owner can incorporate in the Product Backlog
- Attended by Team, Product Owner, ScrumMaster, functional managers, and any other stakeholders
- A demo of what's been built and not a presentation about what's been built:
  - No Power Points allowed!
- Usually lasts 1-2 hours
- Followed by Sprint Retrospective

Sprint Retrospective

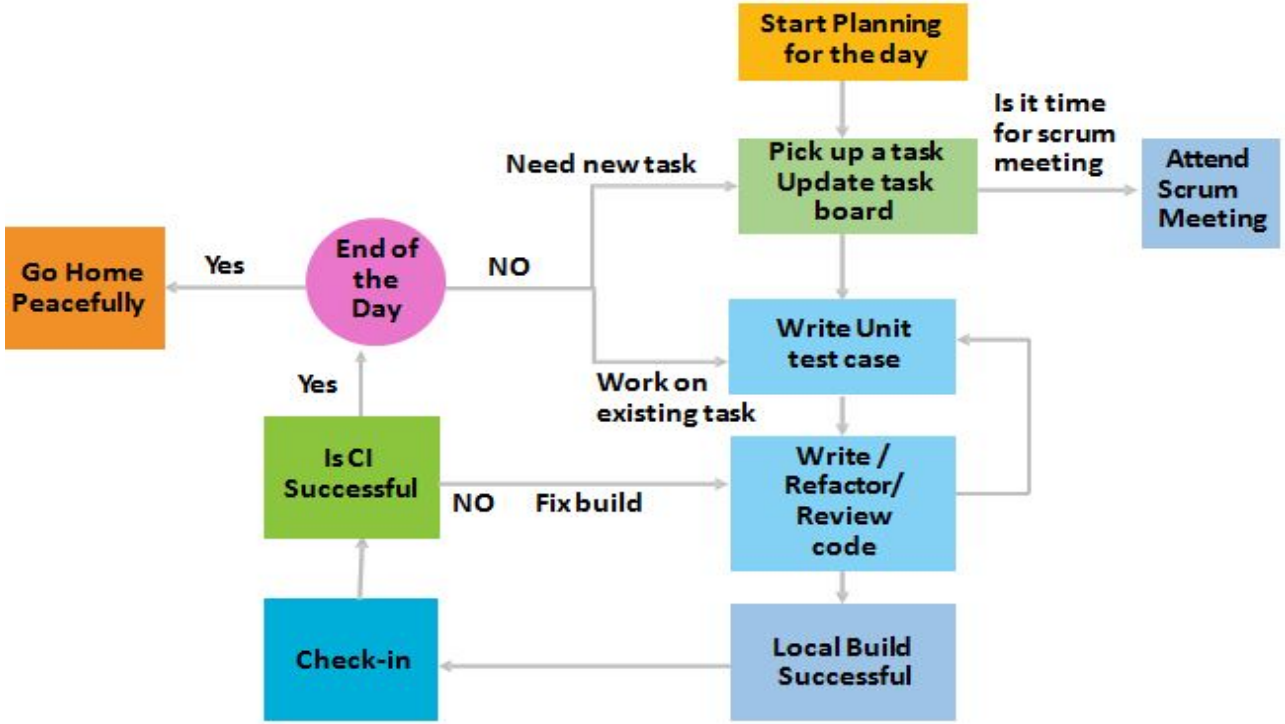
What is it?

- 1-2 hour meeting following each Sprint Demo
- Attended by Product Owner, Team, and ScrumMaster
- Usually a neutral person will be invited in to

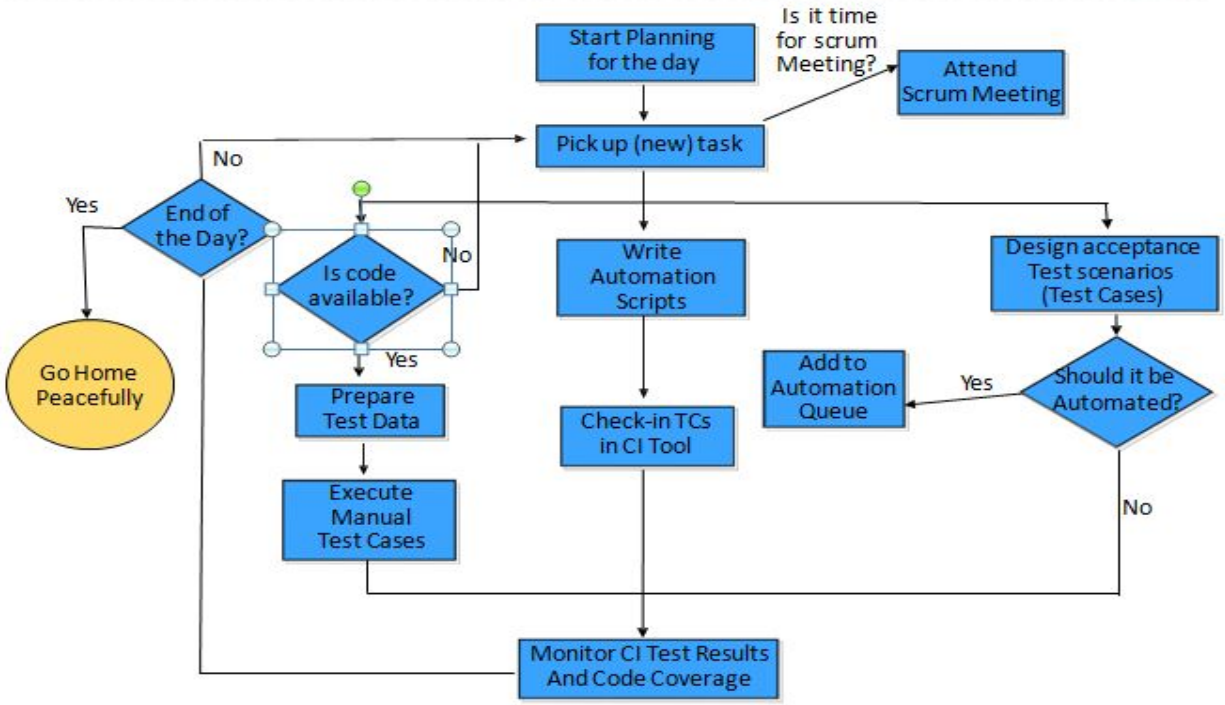
Why does the Retrospective matter?

- Accelerates visibility
- Accelerates action to improve better

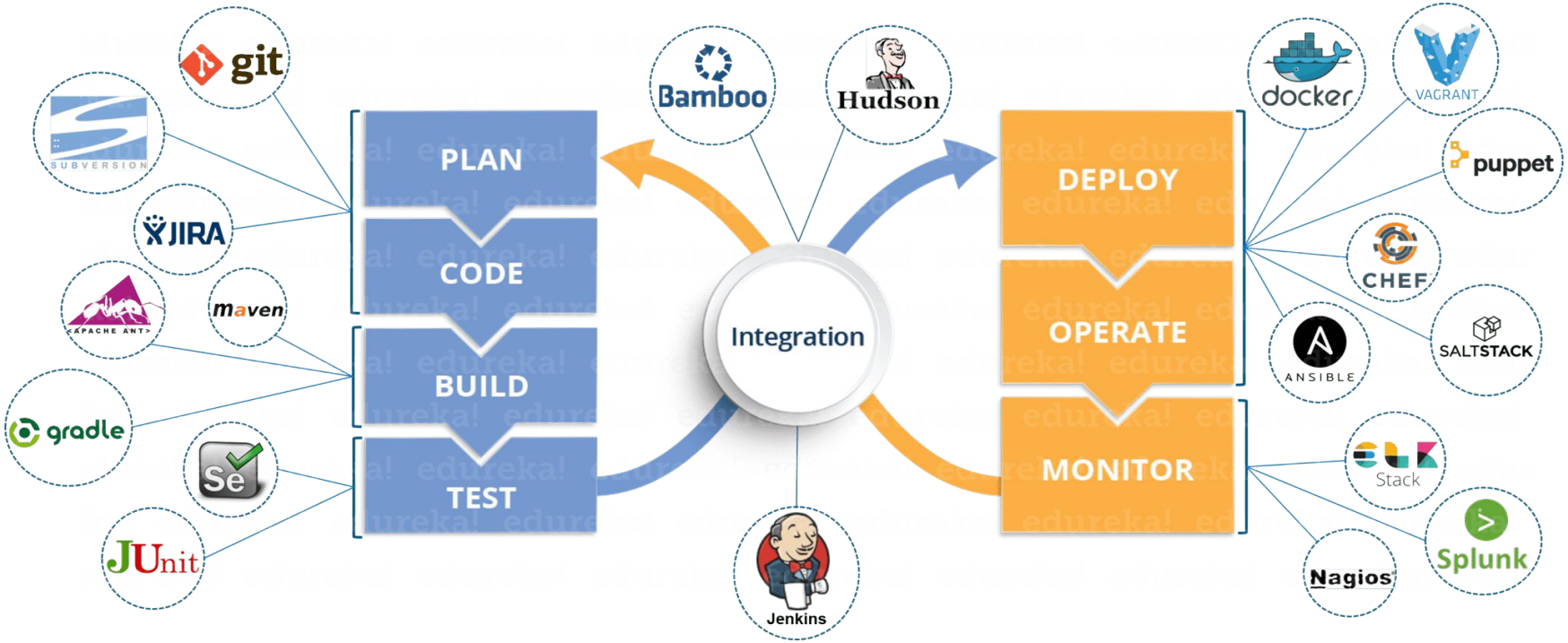
A typical day in the life of a developer



A typical day in the life of a tester



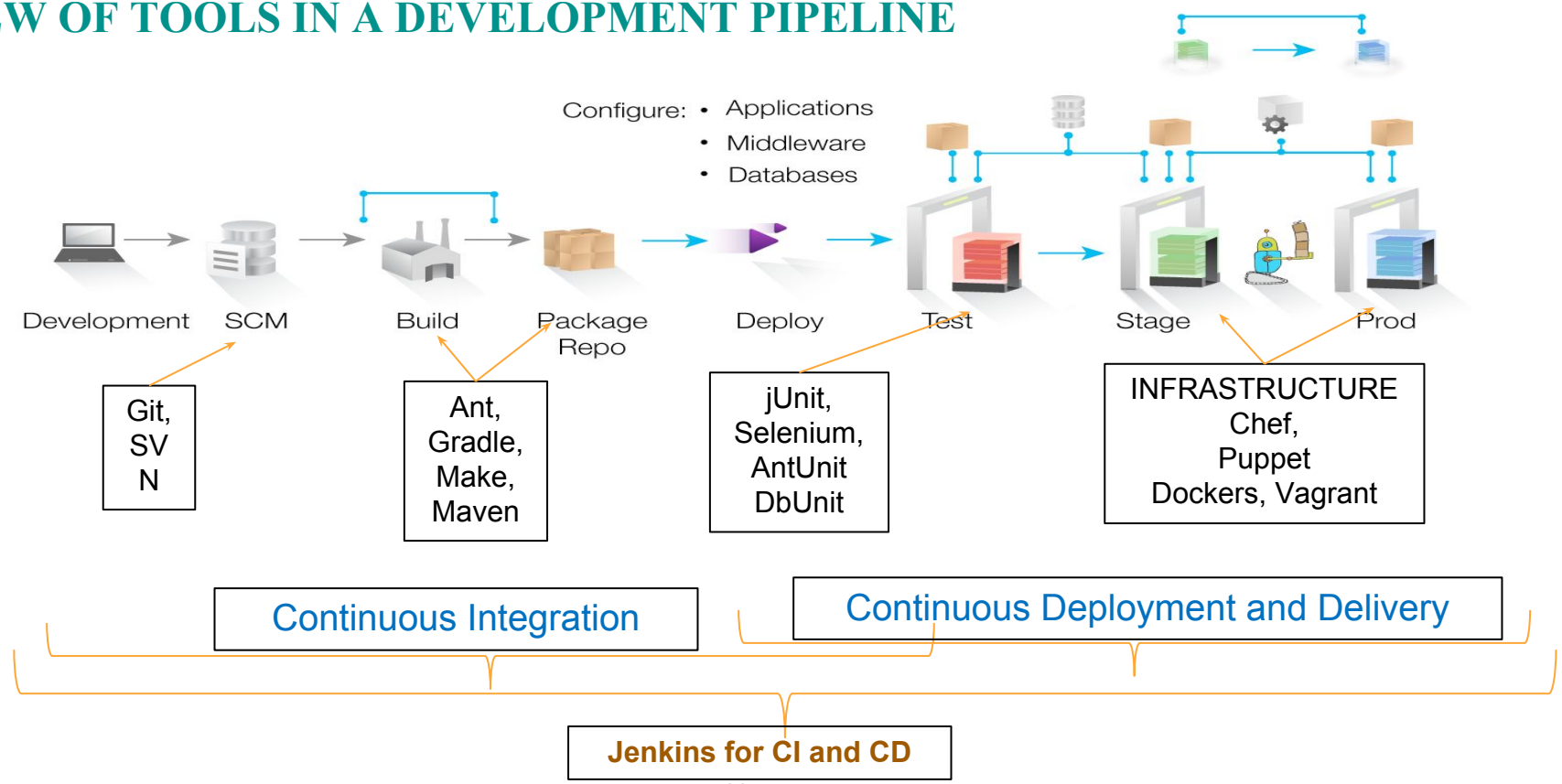
DevOps Tools



Agile & DevOps

Agile	DevOps
Emphasize breaking down barriers between developers and management.	DevOps is about software deployment and operation teams.
Addresses gap between customer requirements and development teams.	Addresses the gap between development and Operation team
Focuses more on functional and non-functional readiness	It focuses operational and business readiness.
Agile development pertains mainly to the way development is thought out by the company.	DevOps emphases on deploying software in the most reliable and safest ways which aren't necessarily always the fastest.
Agile development puts a huge emphasis on training all team members to have varieties of similar and equal skills. So that, when something goes wrong, any team member can get assistance from any member in the absence of the team leader.	DevOps, likes to divide and conquer, spreading the skill set between the development and operation teams. It also maintains consistent communication.
Agile development manages on "sprints. It means that the time table is much shorter (less than a month) and several features are to be produced and released in that period.	DevOps strives for consolidated deadlines and benchmarks with major releases, rather than smaller and more frequent ones.

VIEW OF TOOLS IN A DEVELOPMENT PIPELINE



SUMMARY

In this lesson, you've learned to:

- Introduction to DevOps
- Software Development